

The center of distances of a duplicated base-four achievement set

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Abstract

We determine the center of distances of the achievement set generated by two copies of each number 4^{-p} , $p \geq 1$. It consists exactly of zero and the numbers 4^{-p} . A short digit proof supplies an explicit obstructing point for every other candidate distance and treats finite and arbitrary infinite expansions uniformly. The argument specializes the complementary-tail method of Banakiewicz, Bartoszewicz, Filipczak and Prus-Wiśniowski. It settles this specific semi-fast instance; the general semi-fast classification is outside its scope.

1 Statement and context

For a nonempty set $S \subseteq \mathbb{R}$, its *center of distances* is

$$\mathcal{C}(S) = \{a \geq 0 : \text{for every } x \in S \text{ there is } z \in S \text{ with } |x - z| = a\}.$$

Let $y_{2p-1} = y_{2p} = 4^{-p}$ for $p \in \mathbb{N} = \{1, 2, \dots\}$, and let

$$E = \left\{ \sum_{n \in A} y_n : A \subseteq \mathbb{N} \right\} = \left\{ \sum_{p \geq 1} d_p 4^{-p} : d_p \in \{0, 1, 2\} \right\}. \quad (1)$$

All subsums, including infinite subsums, are permitted. Absolute convergence gives the second description by grouping equal terms.

Theorem 1. *For the set E in (1),*

$$\mathcal{C}(E) = \{0\} \cup \{4^{-p} : p \geq 1\}. \quad (2)$$

The equality concerns all nonnegative real candidate distances.

The lower inclusion is the usual achievement-set inclusion [3, Proposition 2]. The upper inclusion is the issue here. The sequence is semi-fast: each distinct value exceeds the sum of all strictly smaller terms, since

$$2 \sum_{j > p} 4^{-j} = \frac{2}{3} 4^{-p} < 4^{-p}.$$

Nowakowski and Prus-Wiśniowski ask for a general characterization of semi-fast sequences whose center contains no additional distances [3, Problem 7]. Their question does not single out the present instance as unresolved. Theorem 1 determines this instance without providing that general characterization.

2 Explicit exclusion witnesses

Proof of Theorem 1. Write $b_k = 4^{-k}$ and $T_k = 2 \sum_{j>k} b_j = (2/3)b_k$. Legal digit expansions in (1) are unique: if two strings first differ at position k , their leading difference has magnitude at least b_k , while the remaining difference has magnitude at most $T_k < b_k$. This is uniqueness of grouped digits, not of the subset of repeated indices.

For every $k \geq 1$, the two open intervals

$$G_{1,k} = \left(\frac{5}{3}b_k, 2b_k\right), \quad G_{2,k} = \left(\frac{8}{3}b_k, 4b_k\right) \quad (3)$$

are disjoint from E . Indeed, an element with a positive digit before k is at least $4b_k$. If all those digits vanish, its value lies in one of

$$[0, T_k], \quad [b_k, b_k + T_k], \quad [2b_k, 2b_k + T_k],$$

according to its k th digit. These intervals avoid both gaps in (3). At $k = 1$ there is no preceding digit, and the same argument applies; the second gap is above $\max E = 2/3$.

If $a \in \mathcal{C}(E)$, testing $x = 0$ shows $a \in E$. Suppose $a > 0$, let k be its first positive digit position, and write

$$a = d_k b_k + t, \quad t = \sum_{j>k} d_j b_j \in [0, T_k], \quad d_k \in \{1, 2\}.$$

If $d_k = 2$, take $x = b_k \in E$. Then $x - a < 0$ and

$$x + a = 3b_k + t \in \left[3b_k, \frac{11}{3}b_k\right] \subset G_{2,k}.$$

Thus neither possible partner $x - a$ nor $x + a$ belongs to E .

It remains to consider $d_k = 1$. If $t = 0$, then $a = b_k$. If $t > 0$, choose any $p > k$ with $d_p > 0$ and put

$$x = T_k - t + b_p = \sum_{j>k} (2 - d_j) b_j + b_p. \quad (4)$$

This is a legal point of E : its digits through k are zero, its p th digit is $3 - d_p \in \{1, 2\}$, and every other later digit is $2 - d_j \in \{0, 1, 2\}$. In particular, $x \leq T_k < b_k < a$, so $x - a < 0$. On the other hand, $0 < b_p \leq b_k/4$, giving

$$\frac{5}{3}b_k < x + a = \frac{5}{3}b_k + b_p \leq \frac{23}{12}b_k < 2b_k.$$

Hence $x + a \in G_{1,k}$, again excluding both possible partners. The construction uses no termination or periodicity assumption on the tail. We have proved $\mathcal{C}(E) \subseteq \{0\} \cup \{b_k : k \geq 1\}$.

Finally, $0 \in \mathcal{C}(E)$ by taking $z = x$. For any fixed k and any legal digit string of $x \in E$, increase its k th digit by one if it is zero or one, and decrease it by one if it is two. The resulting point $z \in E$ satisfies $|x - z| = b_k$. This proves the reverse inclusion. $\square$

3 Relation to prior results and verification

The witness (4) specializes the complementary-tail construction in the proof of [1, Lemma 3.1], which puts center distances of semi-fast achievement sets among their finite subsums. That general theorem and its method are prior work. In this instance the estimate

$$b_p \leq \frac{1}{4}b_k < b_k - T_k = \frac{1}{3}b_k \quad (p > k)$$

makes the construction exclude every positive later tail, including finite tails. Together with the leading-digit-two witness, this gives the exact upper inclusion in (2). The proof is self-contained and does not require the finite-support theorem as a logical dependency.

The central-Cantor hypothesis of [1, Theorem 3.6] does not hold here. The midpoint $1/3 = \sum_{p \geq 1} 4^{-p}$ belongs to E , whereas a central Cantor set constructed by a nonempty first central deletion omits the midpoint of its convex hull. The separation theorem for multigeometric series in [2, Theorem 4.1] also does not apply by direct substitution. In that paper's normalization, $E(1, 1; 1/4) = 4E$, the total sum is $s = 8/3$, and the finite block set is $F_2 = \{0, 1, 2\}$ with minimum positive separation $\delta(F_2) = 1$. Its hypothesis $2qs < \delta(F_2)$ fails, since $q = 1/4$ gives $2qs = 4/3$. These comparisons identify the scope of the inspected results; they do not establish exhaustive priority. The contribution here is this explicit instance and its exclusion witnesses, using an existing method.

Exact rational computations were used as consistency checks during proof development and review. In particular, the development checks covered all $3^8 = 6561$ legal words of length eight and 2835 distinct rational values represented by prefixes of length zero through three and periodic blocks of length one through four. Such finite coverage does not imply (2); the digit proof supplies the universal conclusion. For broad adjacent context on exact certification of distances in finite ternary configurations, see Zeng, Liu, Ratnavelu and Huat [4]. That reference concerns finite geometric models, not centers of achievement sets.

AI and computational assistance. An OpenAI language model assisted with proof development, exact-rational Python scripts, literature comparison, a separate proof-review pass, and manuscript preparation. Separate model passes are not external peer review. No proof-assistant formalization was performed. The mathematical conclusion follows from the proof above, independently of the diagnostic computations.

References

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